

GUNSAN, Korea, Republic of

WMO: 471400

Lat: 36.0053N

Lon: 126.7614E

Elev: 25

StdP: 101.03

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-6.3	-14.8	1.0	-4.5	-13.0	1.2	-4.2	13.7	-1.0	11.4	0.7	2.0	90	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	31.9	26.4	30.7	25.8	29.5	25.2	27.2	30.3	26.6	29.5	25.9	28.6	3.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.1	28.9	25.7	21.0	28.2	25.1	20.3	27.8	86.5	30.4	83.4	29.8	80.5	28.7	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
10.5	8.8	7.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-10.5	33.9	2.4	1.1	-12.2	34.7	-13.6	35.4	-14.9	36.0	-16.6	36.8		
			-11.4	27.7	2.2	1.0	-12.9	28.4	-14.2	29.0	-15.4	29.5	-17.0	30.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	-0.1	1.6	6.0	11.8	17.3	21.9	25.5	26.3	21.8	15.8	9.0	2.4	(6)
(7)		DBStd	9.60	3.46	3.36	3.33	3.43	2.59	1.89	2.21	2.29	2.45	3.06	3.73	3.72	(7)
(8)		HDD10.0	1002	312	236	132	20	0	0	0	0	2	62	237		(8)
(9)		HDD18.3	2538	571	468	384	199	51	1	0	2	88	280	495		(9)
(10)		CDD10.0	2216	0	1	7	72	227	356	482	504	353	181	32	1	(10)
(11)		CDD18.3	712	0	0	0	2	19	107	223	246	106	9	0	0	(11)
(12)		CDH23.3	4956	0	0	0	8	71	383	1747	2254	477	16	0	0	(12)
(13)		CDH26.7	1284	0	0	0	1	4	39	462	713	64	1	0	0	(13)
(14)	Wind	WSAvg	2.8	2.7	3.0	3.5	3.4	3.1	2.7	2.8	2.6	2.5	2.4	2.7	2.5	(14)
(15)	Precipitation	PrecAvg	1188	33	37	50	87	89	135	251	229	136	47	59	35	(15)
(16)		PrecMax	1769	92	98	189	213	189	326	605	612	290	120	158	65	(16)
(17)		PrecMin	710	7	4	3	13	7	17	67	96	4	1	19	6	(17)
(18)		PrecStd	275	25	28	36	48	45	81	134	108	92	33	38	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	15.4	18.8	24.3	27.1	29.3	33.0	34.0	30.1	25.0	20.5	13.6	(19)
(20)			MCWB	7.7	11.7	13.1	16.9	18.2	21.4	26.2	26.8	23.9	19.1	15.7	9.9	(20)
(21)		2%	DB	8.0	10.8	15.6	21.2	25.2	27.7	31.6	32.4	28.4	23.4	18.0	11.0	(21)
(22)			MCWB	5.5	7.5	10.7	14.7	17.9	21.2	26.1	26.4	23.2	18.3	14.3	8.2	(22)
(23)		5%	DB	6.3	8.4	13.5	19.3	23.8	26.6	30.4	31.4	27.1	21.9	16.3	9.3	(23)
(24)			MCWB	4.0	5.5	9.6	13.7	17.3	21.2	25.8	26.2	22.3	17.4	13.0	6.7	(24)
(25)		10%	DB	4.8	6.6	11.5	17.5	22.3	25.6	29.2	30.3	26.0	20.6	14.7	7.8	(25)
(26)			MCWB	2.6	4.1	8.0	12.7	16.9	21.0	25.4	25.8	21.8	16.6	11.6	5.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	13.1	14.6	18.6	21.6	24.7	28.1	28.5	25.7	21.7	17.2	10.7	(27)
(28)			MCD B	9.9	14.8	17.1	21.7	23.8	26.8	30.6	31.2	28.5	23.7	19.3	12.6	(28)
(29)		2%	WB	6.0	8.0	12.3	16.5	19.8	23.5	27.1	27.6	24.6	19.8	15.2	8.8	(29)
(30)			MCD B	7.8	9.9	14.8	19.7	23.1	25.7	29.9	30.6	27.0	21.7	17.1	10.6	(30)
(31)		5%	WB	4.3	5.9	10.0	14.8	18.7	22.7	26.4	26.9	23.7	18.4	13.6	7.0	(31)
(32)			MCD B	5.9	7.9	12.6	17.9	21.9	24.9	29.2	30.1	25.9	20.8	15.8	8.9	(32)
(33)		10%	WB	2.9	4.5	8.4	13.4	17.8	21.9	25.8	26.3	22.7	17.3	12.0	5.6	(33)
(34)			MCD B	4.6	6.4	10.9	16.4	20.9	24.2	28.4	29.1	25.1	19.6	14.3	7.6	(34)
(35)	Mean Daily Temperature Range	MDBR	6.6	7.0	7.8	8.5	8.2	6.6	5.4	6.0	7.0	8.1	7.5	6.7	(35)	
(36)		5% DB	MCD BR	8.0	9.4	11.2	11.4	10.5	8.4	6.9	7.4	8.0	9.1	9.1	8.6	(36)
(37)			MCWBR	6.1	7.0	7.2	5.9	4.9	3.6	2.9	3.0	3.8	4.8	6.0	6.4	(37)
(38)		5% WB	MCD BR	7.4	8.7	10.1	9.4	8.6	6.1	6.2	6.7	6.6	7.5	7.9	8.1	(38)
(39)			MCWBR	6.2	7.1	7.5	6.0	5.1	3.5	3.2	3.1	3.9	5.0	6.3	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.454	0.493	0.570	0.593	0.611	0.681	0.630	0.574	0.503	0.486	0.472	0.453	(40)	
(41)		taud	1.912	1.851	1.707	1.705	1.722	1.647	1.805	1.942	2.061	1.968	1.947	1.927	(41)	
(42)		Ebn at Noon	691	718	701	716	711	661	692	724	755	718	669	658	(42)	
(43)		Edh at Noon	148	177	223	235	234	252	214	183	154	155	142	138	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.39	3.22	4.36	5.06	5.58	5.09	4.41	4.55	4.17	3.65	2.51	2.09	(44)	
(45)		RadStd	0.17	0.28	0.42	0.40	0.56	0.48	0.62	0.66	0.42	0.29	0.28	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page